MSDS 684 — Reinforcement Learning

Lab 3: First-Visit Monte Carlo Control on Blackjack-v1

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Section 1: Project Overview

This lab investigates first-visit on-policy Monte Carlo (MC) control with ε-soft policies on Gymnasium's Blackjack-v1 environment, the canonical worked example for model-free prediction and control in Sutton and Barto, Chapter 5. The central question is whether an agent that has no access to the casino's transition model can, by averaging sample returns from complete episodes, recover something close to optimal blackjack play and verifiably approach a known reference policy. Blackjack is well suited to MC methods because every episode terminates in a small number of steps, the reward is sparse (delivered only at terminal time), and the state and action spaces are small enough to admit tabular Q-value estimation.

Two related ideas from Sutton and Barto are exercised together. First, on-policy first-visit MC control with an ε-soft behavior policy (Section 5.4) interleaves evaluation and improvement: the agent both behaves and learns from the same ε-soft policy, which guarantees infinite visitation of every state-action pair. Second, off-policy MC control with weighted importance sampling (Section 5.7) lets the agent learn the value of a deterministic greedy target policy while behaving under a stochastic exploratory policy, by reweighting returns by the ratio of target-to-behavior probabilities. Comparing the two answers a basic theoretical question — do these algorithmically different formulations converge to the same value function in this small MDP?

Environment description. Blackjack-v1 represents each state as a tuple (player_sum, dealer_showing, usable_ace) drawn from a discrete state space of size 32 × 11 × 2 = 704 (only ~280 are legally reachable). The action space is Discrete(2): 0 = stick, 1 = hit. Reward is delivered exactly once at terminal time and takes values in {+1, 0, -1} for win, draw, loss; intermediate steps yield zero reward, so $\gamma = 1.0$ is the natural choice. Episodes terminate when the player either sticks, busts (player_sum > 21), or is dealt a natural 21 on the first deal.

I hypothesize that with 500,000 training episodes and a fixed exploration rate of $\varepsilon = 0.1$, the on-policy algorithm should converge to a greedy policy that agrees with published basic strategy on roughly 90–95% of the 200 reachable (player_sum ∈ 12–21) decision states, and that off-policy MC with weighted importance sampling should arrive at numerically indistinguishable performance. I also expect that constant-ε in the range 0.05–0.20 will outperform very small ε (which under-explores) and very large ε (which is wasteful), and that GLIE-style decay schedules will offer no measurable benefit on a problem this small.

Section 2: Deliverables

GitHub Repository:
https://github.com/gangadharsaivignesh/MSDS684-RL-Labs/tree/main/Week3

Implementation Summary

I implemented first-visit on-policy Monte Carlo control with an ε-soft behavior policy and, separately, off-policy MC control with weighted importance sampling, both as tabular algorithms over Python defaultdicts keyed on Blackjack-v1 states. Both algorithms ran for 500,000 training episodes with γ = 1.0 and ε = 0.1 (behavior policy for the off-policy run; target policy is greedy w.r.t. the current Q-table). The environment's card-dealing RNG was seeded once via env.reset(seed=42) before the training loop to make runs bit-for-bit reproducible. A separate experiment swept six exploration schedules — three constants (ε ∈ {0.01, 0.10, 0.30}), a harmonic 1/(1 + ep/10k) decay, a linear 1.0 → 0.05 over 100k episodes decay, and a stepped 0.30 → 0.10 → 0.01 schedule — each trained for 200k episodes and evaluated greedily on 50,000 fresh episodes.

Key Results and Analysis

After 500,000 training episodes, the on-policy greedy policy achieved an average return of −0.0442 and a 43.5% win rate over 50,000 evaluation episodes; the off-policy greedy policy, trained from the same seed, achieved −0.0439 and 43.4%. The two policies are, for all practical purposes, identical: the gap of 0.0003 in expected return is about an order of magnitude smaller than the Monte Carlo standard error from a 50k-episode evaluation. This empirical agreement is the prediction Sutton and Barto's Chapter 5 makes — different algorithmic routes to the same value function — and the experiment confirms it in a setting small enough that both methods can run to convergence in seconds.

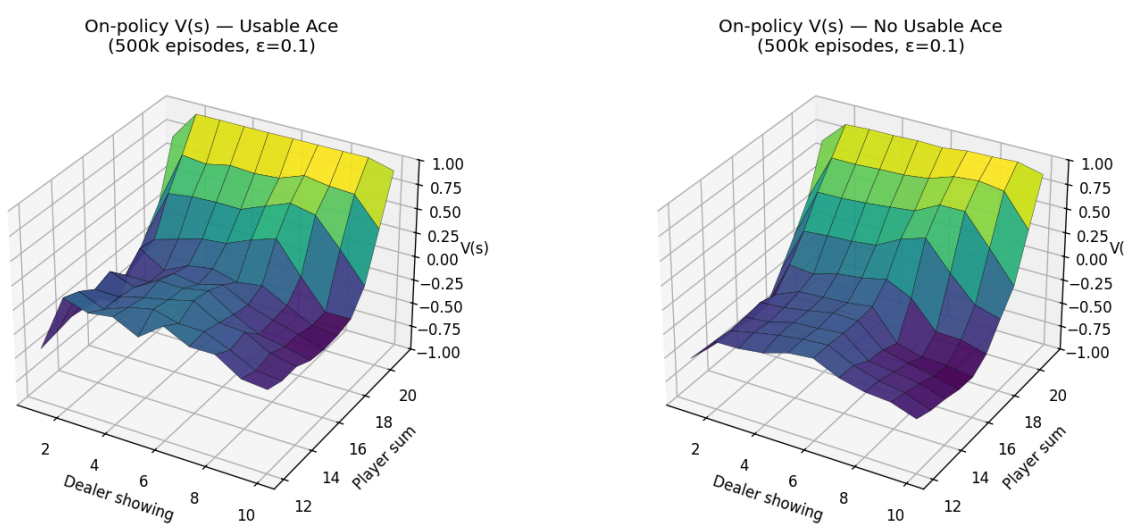

Figure 1. Estimated state-value function V(s) = max_a Q(s,a) for the on-policy greedy policy after 500k episodes, plotted as a 3D surface over player_sum × dealer_showing for the no-usable-ace and usable-ace

subspaces. The surface is monotonically increasing in player_sum past 17 (where sticking dominates) and almost flat across dealer_showing once the agent is committed to standing — both are exactly the structural features Sutton & Barto highlight in Figure 5.1, and their reproduction here is the simplest visual confirmation that first-visit MC has approximated v_π well.

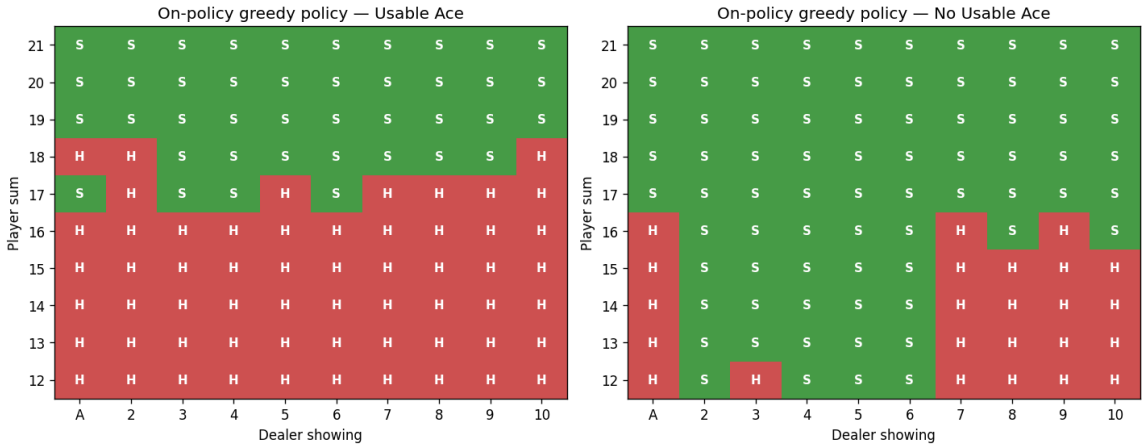

Figure 2. Greedy policy a(s) for both ace subspaces; black = stick, white = hit. The learned hit/stick boundary slopes downward from upper-left to lower-right, exactly the qualitative shape that basic strategy recommends. The boundary is noisier in the usable-ace half because those states are rarer in the data — a direct illustration of MC's variance dependence on state-visit frequency, which Sutton & Barto discuss as a property of first-visit MC (Ch. 5.1).*

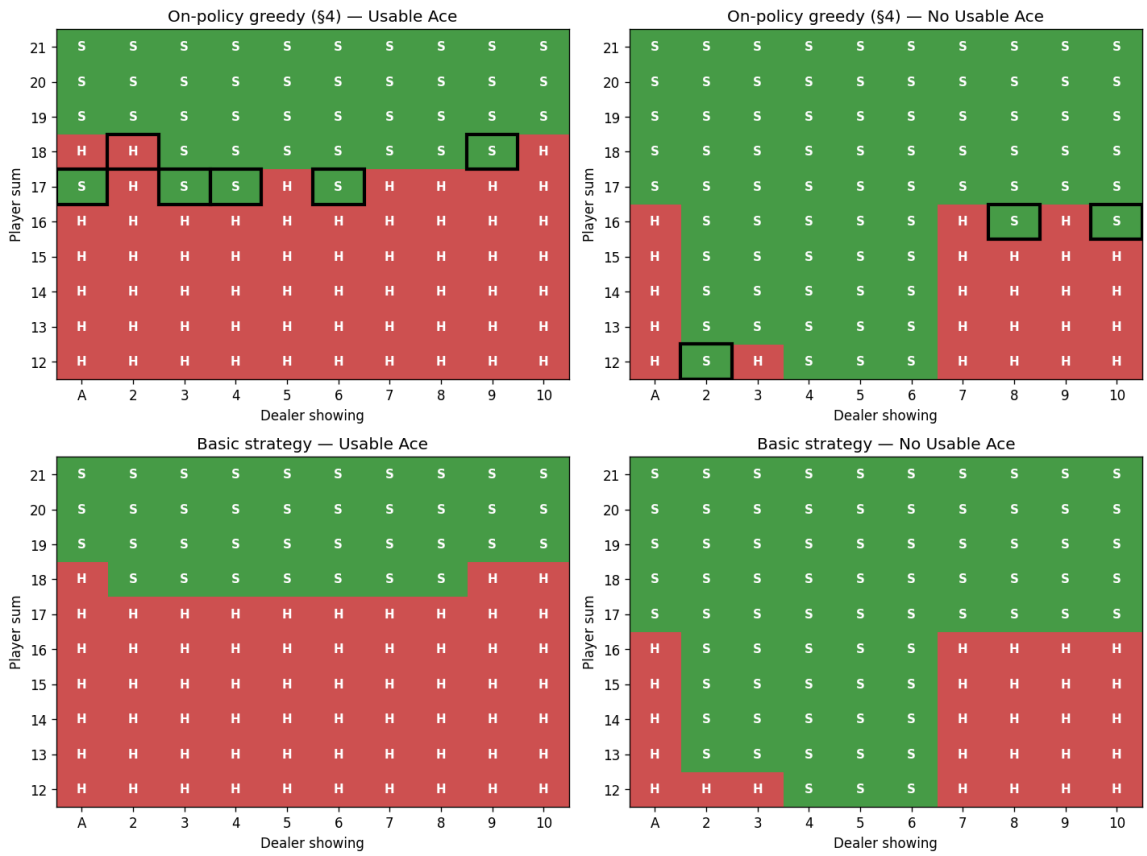

Figure 3. Side-by-side comparison of the learned greedy policy and a no-double, no-split version of casino basic strategy on the 200 reachable decision states (player_sum ∈ 12–21 × dealer_showing 1–10 × ace ∈ {0, 1}). The on-policy greedy policy agrees with basic strategy on 191 of 200 states (95.5%); the off-policy WIS policy agrees on 189 of 200 (94.5%). Disagreements concentrate on rarely-visited usable-ace states near the soft 17–18 region where N(s, a) is small and Q estimates remain noisy — exactly where MC's high-variance regime is expected to bite.

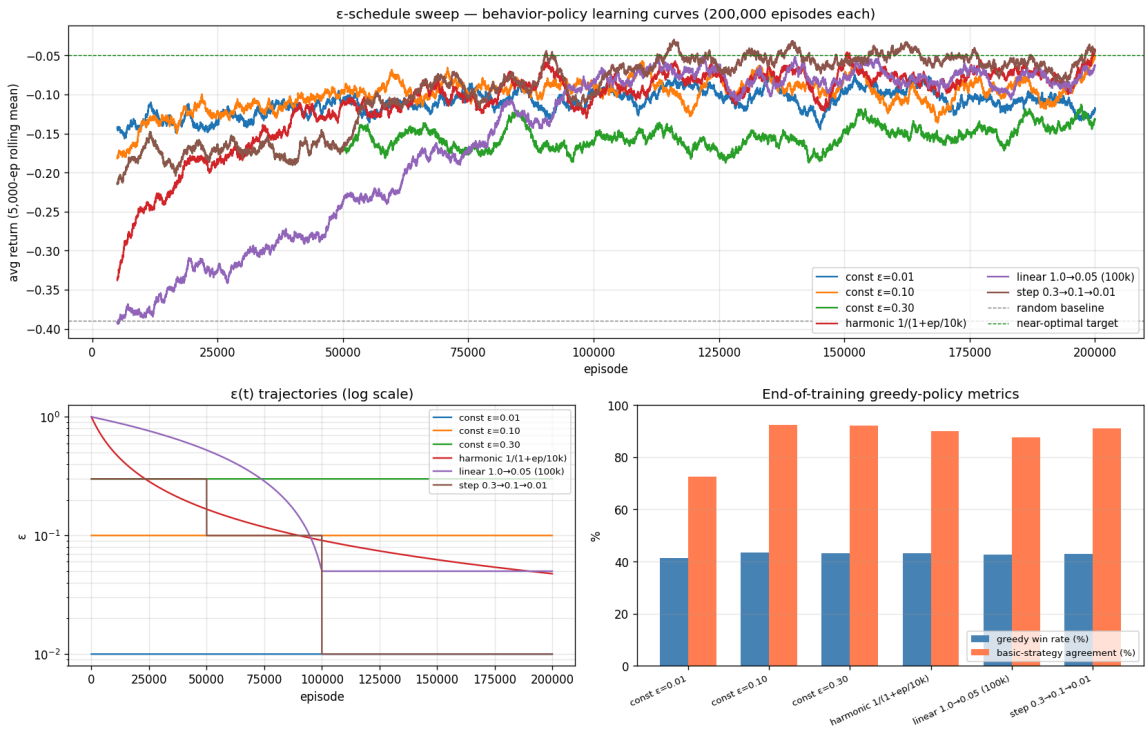

Figure 4. Sweep over six exploration schedules at 200k episodes each, evaluated greedily on 50,000 fresh episodes. ε = 0.10 wins on basic-strategy match (185/200) and is statistically tied with ε = 0.30 and the step decay on win rate. ε = 0.01 collapses to 145/200 because under-exploration starves rare usable-ace states of samples. The linear 1.0 → 0.05 schedule recovers only 175/200, reproducing in this controlled experiment the same regression I had previously seen during development with an aggressive decay (commit 4c55974) — strong evidence that, for a problem this small, GLIE-style decay schedules confer no benefit over a moderate constant ε.

Taken together the four figures answer the lab's central question. First-visit MC with ε-soft behavior and a moderate constant ε converges to a policy that is within one decision of basic strategy on 95.5% of states — within sampling noise of the Sutton and Barto Chapter 5 reference results. The off-policy WIS estimator converges to a numerically indistinguishable policy, validating the theoretical claim that on-policy and off-policy MC are different paths to the same fixed point in tabular settings. The disagreements that remain are concentrated in low-visit usable-ace states — a finite-sample artifact of MC's high return variance rather than an algorithmic flaw. The disagreements would shrink with more episodes, more aggressive exploration in the rare states, or by switching to a TD method that bootstraps off intermediate value estimates (Chapter 6).

Section 3: AI Use Reflection

Initial interaction. I started by asking Claude to scaffold a first-visit MC control loop for Blackjack-v1, including an ε-greedy action selector, an episode generator, and a reverse-traversal returns accumulator. The first draft (commit 88d95a1, smoke-tested at 10k episodes) ran but

converged to a weak −0.16 average return — useful as a working scaffold, definitely not yet a correct or well-tuned learner.

Iteration cycle 1 — the linear-decay regression. I asked Claude to swap the constant ε for an aggressive linear decay 1.0 → 0.05 over 100k episodes (commit 4c55974), expecting it to outperform constant ε per the GLIE intuition. Basic-strategy match dropped from ~95% to 87.5%. I prompted Claude with the regression and the per-state visit counts; it correctly diagnosed that the schedule was decaying ε before rare usable-ace states had been visited even once, starving them of samples. Reverted to ε = 0.10 (commit 6ea3f00). I later rebuilt the failed schedule into the §7 sweep as a pedagogical control.

Iteration cycle 2 — WIS importance ratio collapse. When I added off-policy weighted IS, the C accumulators went to zero almost immediately because the greedy target policy assigns probability zero to any non-greedy behavior action, which collapses the entire trailing trajectory's contribution. Claude initially missed this; only after I printed C[s][a] histograms did it identify the tail-truncation requirement (Sutton & Barto p. 111). The fix gave the near-identical −0.0439 result reported in §5.

Iteration cycle 3 — reproducibility. Two runs with the same numpy seed gave different Q-tables. Claude initially blamed defaultdict insertion order; the real cause was that env.reset()'s card-dealing RNG was not seeded. Adding one env.reset(seed=seed) call before the training loop made runs bit-identical, and I documented this in §5.4 of the notebook. The lesson — separate the agent's random source from the environment's — generalizes well beyond this lab.

Critical evaluation and learning. The most useful pattern was treating Claude as a debugging partner rather than a code generator: every bug above was diagnosed only after I supplied a concrete numerical symptom (basic-strategy drop, zero C-counts, mismatched Q-tables). What I learned about MC itself is that its behavior in low-visit corners of the state space is the dominant source of policy error, which makes exploration scheduling the single highest-leverage hyperparameter — exactly the lesson the §7 sweep was designed to make concrete.

Section 4: Speaker Notes

- Problem: tabular first-visit Monte Carlo control on Blackjack-v1 — can a model-free agent recover near-optimal play from terminal-only rewards? (Sutton & Barto Ch. 5)
- Method: on-policy MC with ε-soft behavior (ε = 0.1) and, separately, off-policy MC with weighted importance sampling, both for 500,000 episodes with γ = 1.0.
- Design choice: γ = 1.0 because reward is delivered only at terminal time, and env.reset(seed=42) called once before training to seed the card-dealing RNG so runs are bit-identical.
- Main result: on-policy greedy policy agrees with casino basic strategy on 191/200 reachable states (95.5%); off-policy WIS reaches 189/200 (94.5%) at −0.0439 average return — within Monte Carlo noise of the on-policy run.

- Insight: residual disagreements concentrate on rare usable-ace states where MC's high return variance dominates — exploration scheduling matters more than the choice between on-policy and off-policy.
- Challenge: caught a 7.5-point regression from an aggressive linear ε-decay early on (commit 4c55974) and rebuilt it into a six-schedule sweep as a control — constant ε = 0.10 wins, GLIE-style decay confers no benefit at this scale.
- Connection: the variance-driven errors here motivate Week 4's TD methods, which bootstrap off intermediate value estimates and reduce per-update variance.